Akshar Raikanti

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EDUCATION

Purdue University **West Lafayette, IN**
B.S., Computer Science **December 2026**
B.S., Artificial Intelligence **GPA: 3.62**
Relevant Courses: Analysis of Algorithms, Systems Programming, Artificial Intelligence, Data Structures and Algorithms

Skills: Python, Java, C, C++, C#/.NET, SQL, R, JavaScript, TensorFlow, PyTorch, Relational Databases, NoSQL, Azure, Google Cloud (GCP), AWS, LangChain, LangGraph, Agentic AI, RAG, Vector Database Search, ML Training, Docker, CI/CD, OpenShift, Jira, Apache, Git

PROFESSIONAL EXPERIENCE

Tesla **Fremont, CA**
Software Engineer Intern **September 2025 – Present**

- Building a Health Check AI Agent that correlates logs, runs post deployment checks to verify application; detects early warning patterns using RAG on historical data, catching potential issues with 85% accuracy.
- Architected a Kubernetes native observability stack on (AWS, Prometheus, Splunk) for material flow services, reducing time to detect for support engineers by ~10%, supporting factory-critical systems by resolving integrations and workflows.
- Designing an optimized MySQL schema aligned to query patterns with partitioning enabled, to allow for efficient execution and consistent low latency reads/writes at scale, aligned to Go/Kafka services linking ERP and manufacturing execution.

Textron **Hunt Valley, MD**
Agentic AI Software Engineer Intern **August 2025 – September 2025**

- Built an Agentic AI RAG workflow in C# + Python that ingests docs; tokenizes, generates and indexes OpenAI embeddings; decision agent reaches the right team with context leading to 40% fewer misroutes using Microsoft Semantic Kernel.
- Trained a bid/no-bid decision engine (RAG on Textron data via FAISS + GPT-4o) that 3x screening throughput and cut manual review time by 50%, while including win-probability scoring on high priority opportunities.
- Collaborated with Microsoft to build a Databricks ingestion/embedding pipeline using Logic Apps for orchestration and Microsoft AI Foundry for model lifecycle integrated with Azure Data Factory to deliver low-latency agent knowledge lookup.

Kohl's **San Francisco, CA**
Software Engineer Intern **June 2025 – August 2025**

- Engineered a scalable high-performance backend for an ordering platform with RESTful API, leveraging Java Spring Boot, MongoDB, Google Cloud Spanner cloud databases, achieving 10% increase in responsiveness, 5% reduction in load times.
- Developed intuitive frontend interfaces using Angular (TypeScript), enhancing Kohl's order management; modernized CI/CD pipelines using Auto DevOps and container orchestration via Red Hat OpenShift (Kubernetes) accelerating deploys by 5%.

TransSIGHT **San Francisco, CA**
Data Engineering and Machine Learning Intern **June 2024 – August 2024**

- Built and deployed predictive machine learning transit-demand models (Random Forest, XGBoost; scikit-learn, PyTorch) and designed an automated end-to-end data pipeline using Apache Airflow, AWS S3/Lambda, raising forecast accuracy by 20%.

PROJECTS

Agentic AI Workflow

- Engineered an agentic Web Search + Retrieval-Augmented Generation (RAG) system on Nvidia NIM (Llama-3.1-8B) using AIQ Toolkit, LangGraph (ReAct/tool-calling), and LangChain; registered a custom Tavily Internet Search Tool and a local-filesystem RAG pipeline (document chunking, embeddings, vector search via FAISS) to deliver fast multi sourced answers to user requests.
- Productionized the agent as a FastAPI + LangGraph microservice (MCP-compatible) with Redis caching, request dedup, and streaming.

Local LLM

- Built and deployed a local LLM by fine-tuning a T5 Seq2Seq model on Q&A using Hugging Face Transformers and PyTorch; evaluated with ROUGE metrics, optimized via BitsAndBytes 8-bit quantization, benchmarked latency with custom scripts; served with a secured FastAPI.

Pantry Tracker App

- Built a responsive web app using React and Next.js for real-time pantry item tracking; implemented secure user authentication via Firebase and automated item addition using computer vision with OpenCV.

RESEARCH EXPERIENCE

BASF – Agricultural Chemical Solutions **West Lafayette, IN**
Machine Learning Research Intern **August 2023 – May 2024**

- Developed statistical models and predictive analytics using Python, scikit-learn, XGBoost, and Tableau, enhancing competitor market-share estimation accuracy and delivering actionable benchmarking insights to senior management leading to stronger business strategy decisions.